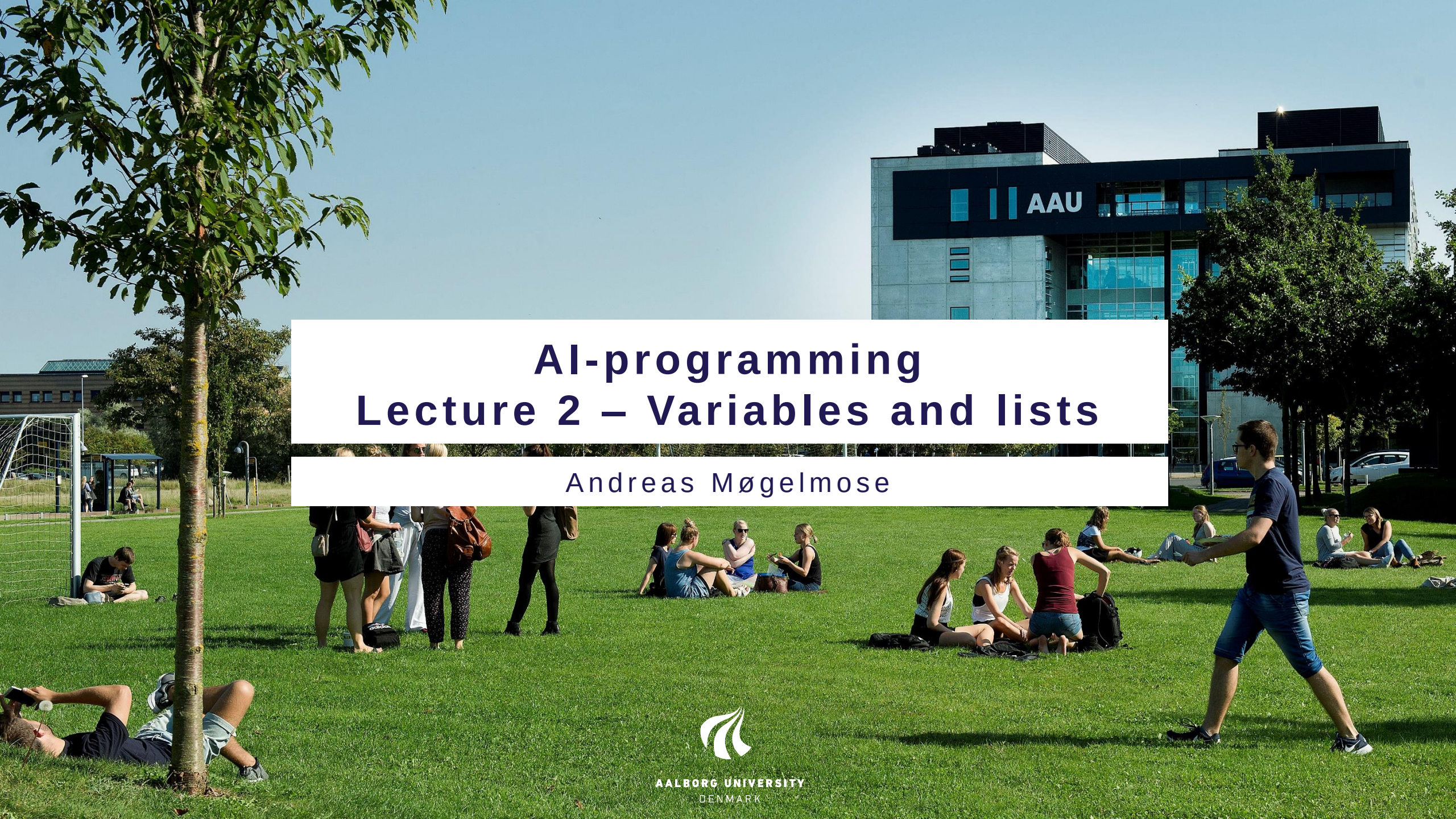


AI-programming
Lecture 2 – Variables and lists

Andreas Møgelmose

AALBORG UNIVERSITY
DENMARK

Agenda

- What are variables, and how to use them?
- What are lists
- Math



Recap: Functions


```
Debug.Log("Hello world");

move(7, 2);

QuitProgram();
```



Agenda

- **What are variables, and how to use them?**
- What are lists
- Math





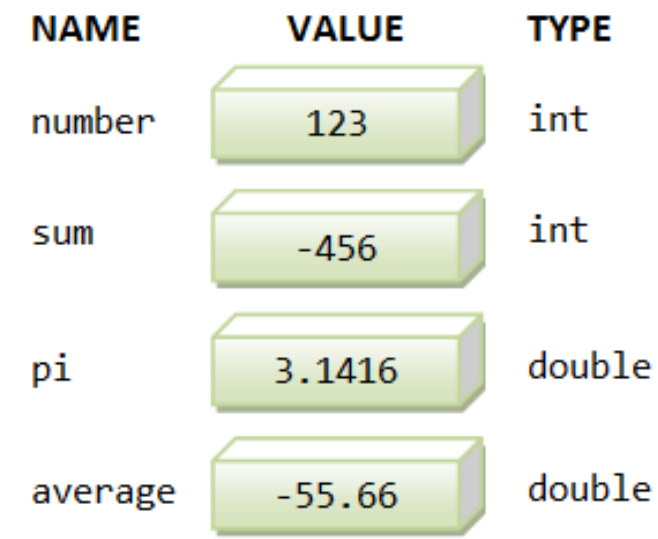
What are variables?

- Just a place to store things temporarily!
- Technically: A named piece of computer memory (RAM)

- Examples:
  - Intermediate results of calculations
  - Text to show the user
  - 3D graphics

- Almost everything in your programs

Variable properties



A variable has a ***name***, stores a ***value*** of the declared ***type***

refers to a

- Name (identifier):
  - Case-sensitive
  - Letters, numbers, and underscores
  - Cannot start with a number.
  - Note: No spaces!
  - No reserved words (if, else, for, etc.)
- Value: Whatever you want to store
- Type: The overall type of thing you want to store (a number, some text, a picture, etc) – Python chooses this automatically

Variable names

- Names – valid or invalid, good or bad?

_456

Hello,hello

number

heading5

xyz

armPosition

time_

123

2names

begin

HELLO

_____

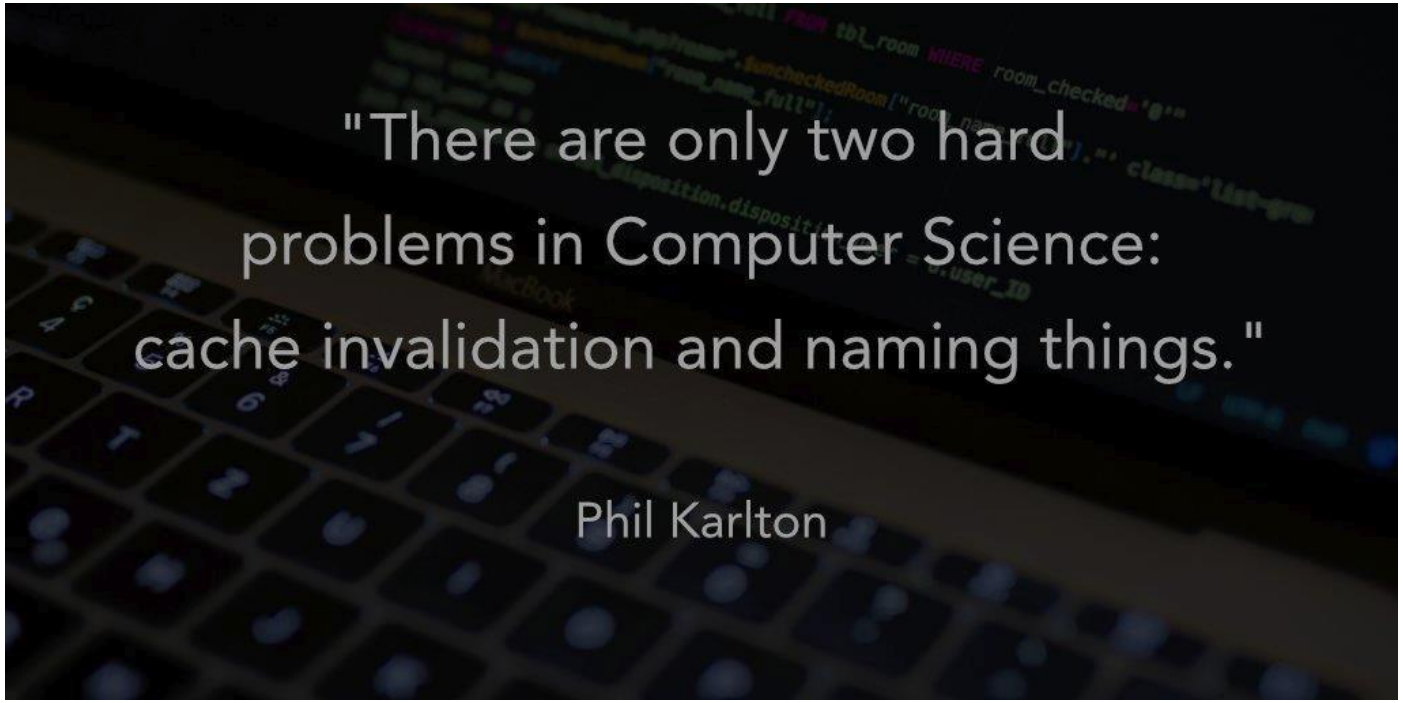
userName

UsErNaMe

topContent

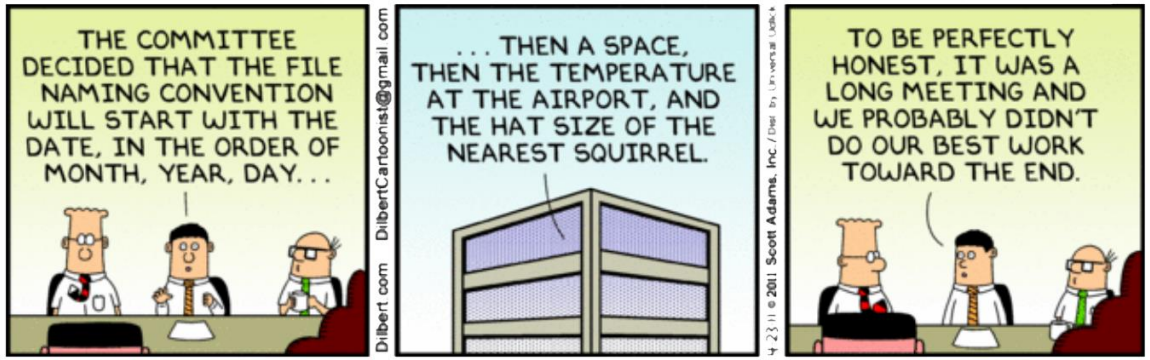
playerScore

Variable names



Naming conventions

- underscore_as_separator
- camelCase
- PascalCase (also sometimes called camel case)



Naming Conventions

- Look up conventions for your language – or do whatever but be consistent!
- For Python: https://kortlink.dk/2mk3p

Data types

- The most important data types (in Python)
  - Strings
  - Integers
  - Floats
  - None

Examples:

```
name = “Andreas”
name = ‘Andreas’

age = 16


averageAge = 8.452
```

```
>>> name = None
>>> type(name)
<class 'NoneType'>
>>> name = "Andreas"
>>> type(name)
<class 'str'>
```

String tips and tricks

```
>>> print("Jeg\n\thedder\n\t\tAndreas")
Jeg
        hedder
                Andreas
```

```
name = "Andreas møgelmose"
print(name.title()) # Andreas Møgelmose
print(name.upper()) # ANDREAS MØGELMOSE
print(name.lower()) # andreas møgelmose
```

```
>>> print('Andreas's hus er flot!')
  File "<stdin>", line 1
    print('Andreas's hus er flot!')
          ^^^^^^^^^^
SyntaxError: invalid syntax. Perhaps you forgot a comma?
```

```
player = "Andreas"
score = 100
message = f"Congratulations {player}, you scored {score} points!"
# message contains Congratulations Andreas, you scored 100 points!
```



Number tips and tricks


```
>>> 5*2
10
>>> 5**2
25
>>> 5/2
2.5
>>> 5//2
2
>>> 5%2
1
```

Remember this one!!

```
>>> a, b, c = 3, 2, 1
>>> print(f"a: {a}, b: {b}, c: {c}")
a: 3, b: 2, c: 1
```



Bonus math operators


```
>>> age = 10
>>> age + 10
20
>>> print(age)
10
```

```
>>> age = 10
>>> age = age+10
>>> print(age)
20
>>> age += 10
>>> print(age)
30
```

```
>>> age = 10
>>> age *= 5
>>> print(age)
50
```



Agenda

- What are variables, and how to use them?
- **What are lists?**
- Math



Lists


```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Dora']
>>> print(names[1])
Bob
```

Zero-indexed!!

```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Dora']
>>> names[3] = "Andreas"
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Andreas']
```



Lists


```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> names.append("Eric")
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Dora', 'Eric']
```

```
>>> names = []
>>> names.append("Alice")
>>> print(names)
['Alice']
```



Lists


```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> print(names.pop())
Dora
>>> print(names)
['Alice', 'Bob', 'Charlie']
```

```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> names.remove("Bob")
>>> names.remove("charlie")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: list.remove(x): x not in list
>>> print(names)
['Alice', 'Charlie', 'Dora']
```

Lists


```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> print(names.pop())
Dora
>>> print(names)
['Alice', 'Bob', 'Charlie']
```

```
>>> names = ["Alice", "Bob", "Charlie", "Dora"]
>>> names.remove("Bob")
>>> names.remove("charlie")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: list.remove(x): x not in list
>>> print(names)
['Alice', 'Charlie', 'Dora']
```

Lists


```
>>> names = ["Bob", "Dora", "Charlie", "Alice"]
>>> names.sort()
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Dora']
```

```
>>> names = ["Bob", "Dora", "Charlie", "Alice"]
>>> names = sorted(names)
>>> print(names)
['Alice', 'Bob', 'Charlie', 'Dora']
```



Agenda

- What are variables, and how to use them?
- What are lists?
- **Math**



AALBORG UNIVERSITY
DENMARK

Math


```
>>> import math
```

Trigonometric functions

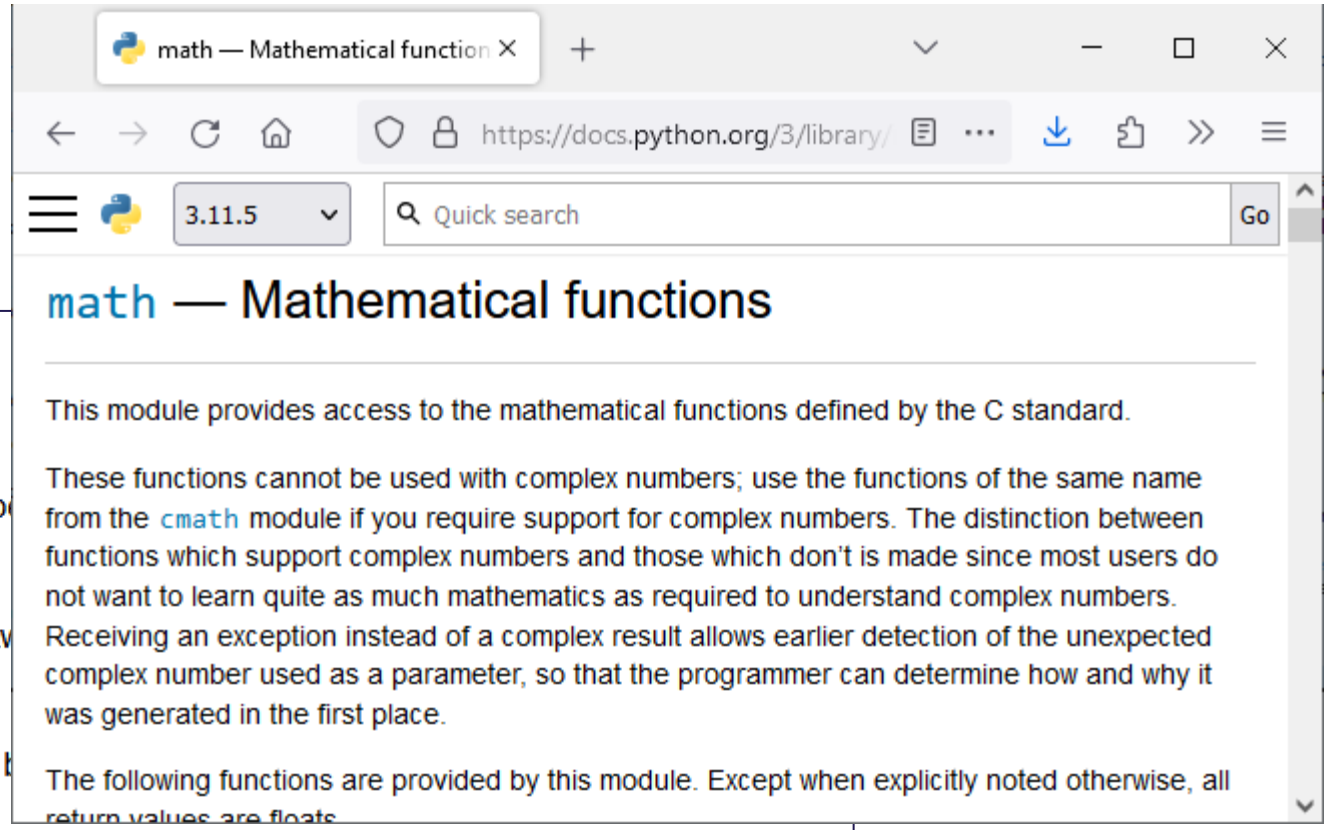
math.acos(*x*)
Return the arc cosine of *x*, in radians. The result is b

math.asin(*x*)
Return the arc sine of *x*, in radians. The result is betv

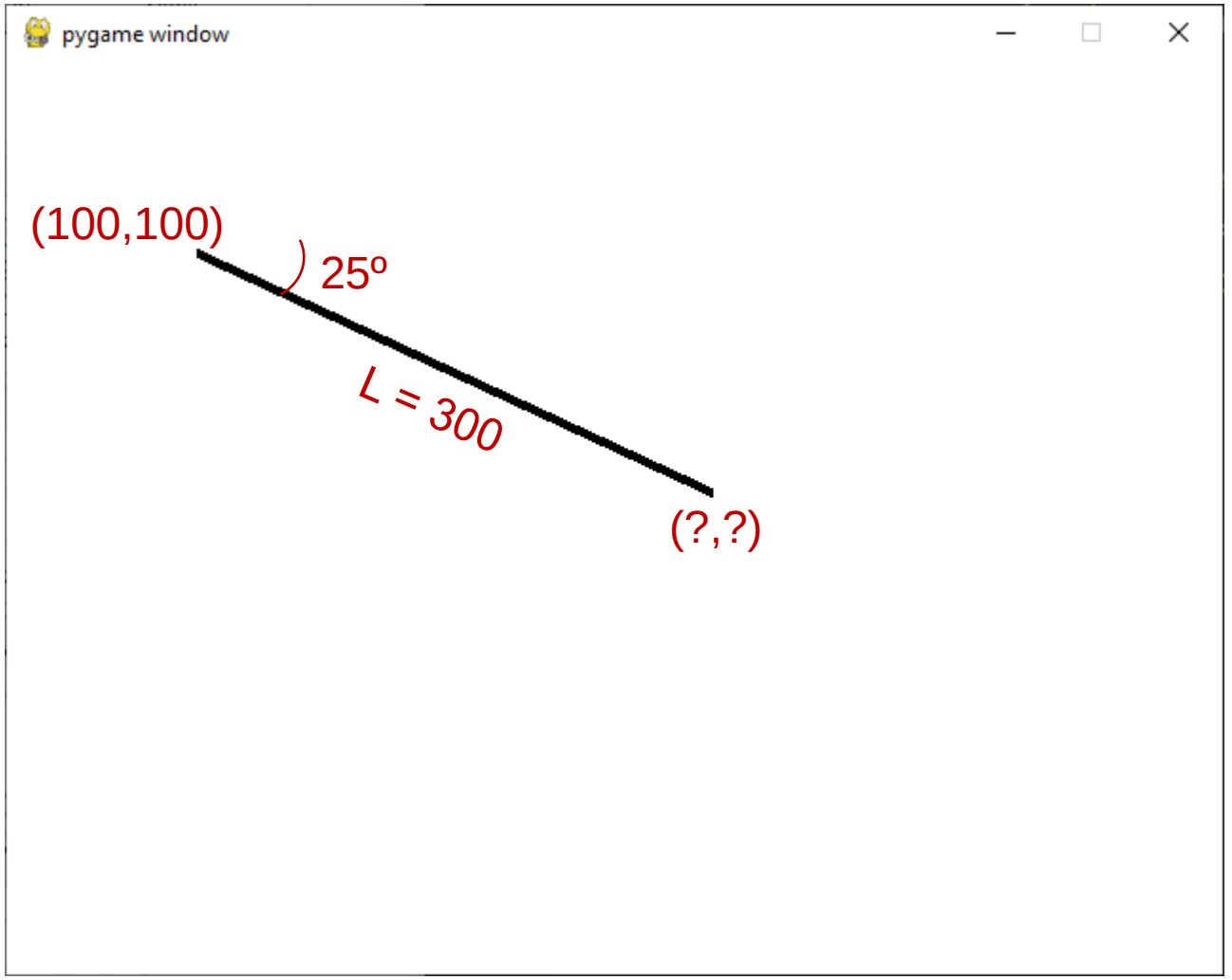
math.atan(*x*)
Return the arc tangent of *x*, in radians. The result is b

math.atan2(*y*, *x*)
Return `atan(y / x)`, in radians. The result is between `-pi` and `pi`. The vector in the plane from the origin to point `(x, y)` makes this angle with the positive X axis. The point of `atan2()` is that the signs of both inputs are known to it, so it can compute the correct quadrant for the angle. For example, `atan(1)` and `atan2(1, 1)` are both `pi/4`, but `atan2(-1, -1)` is `-3*pi/4`.

math.cos(*x*)

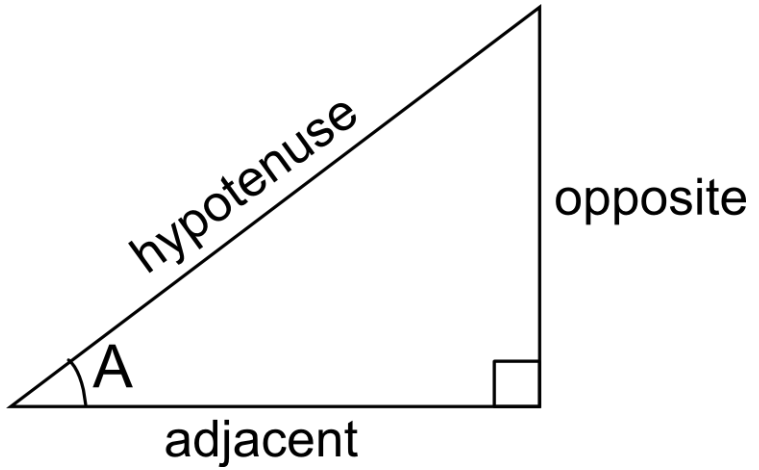


Drawing a line at an angle

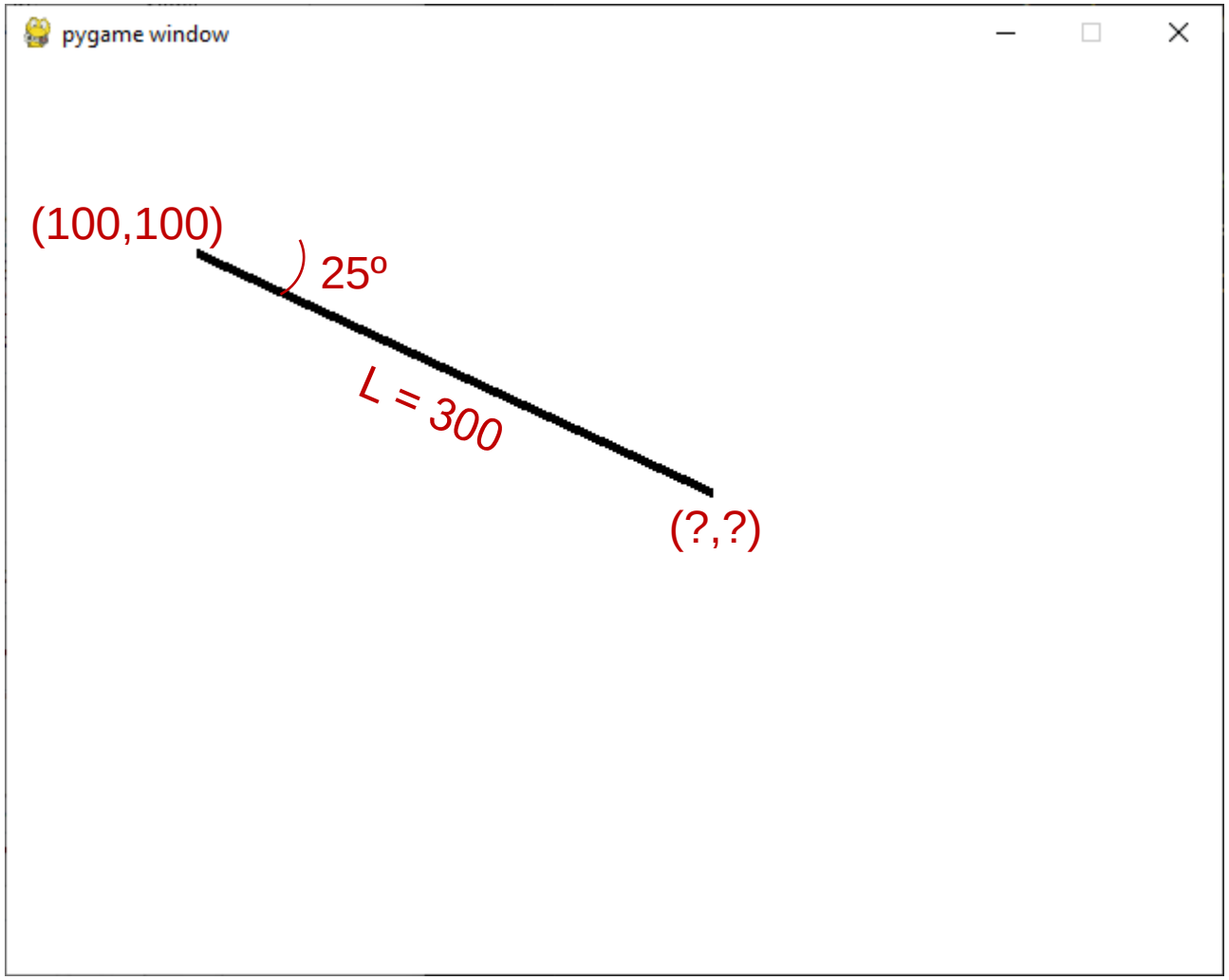


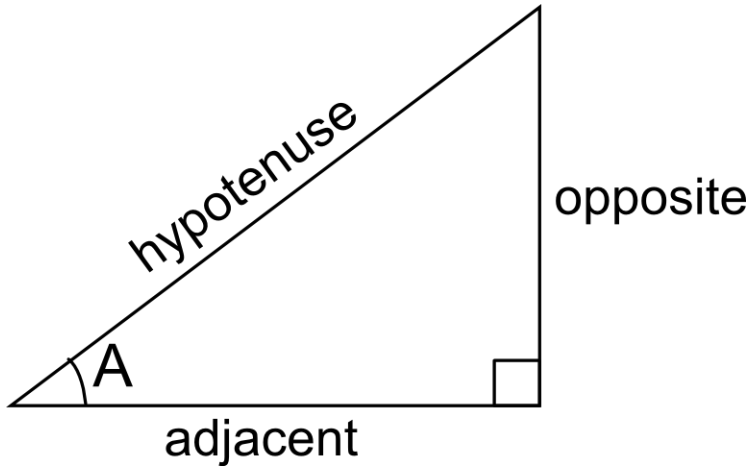
Trigonometry

- Rules for right-angle triangles:
  - $\sin(A) = \frac{opposite}{hypotenuse}$
  - $\cos(A) = \frac{adjacent}{hypotenuse}$
  - $\tan(A) = \frac{opposite}{adjacent}$
  - $adjacent = \cos(A) * hypotenuse$
  - $opposite = \sin(A) * hypotenuse$

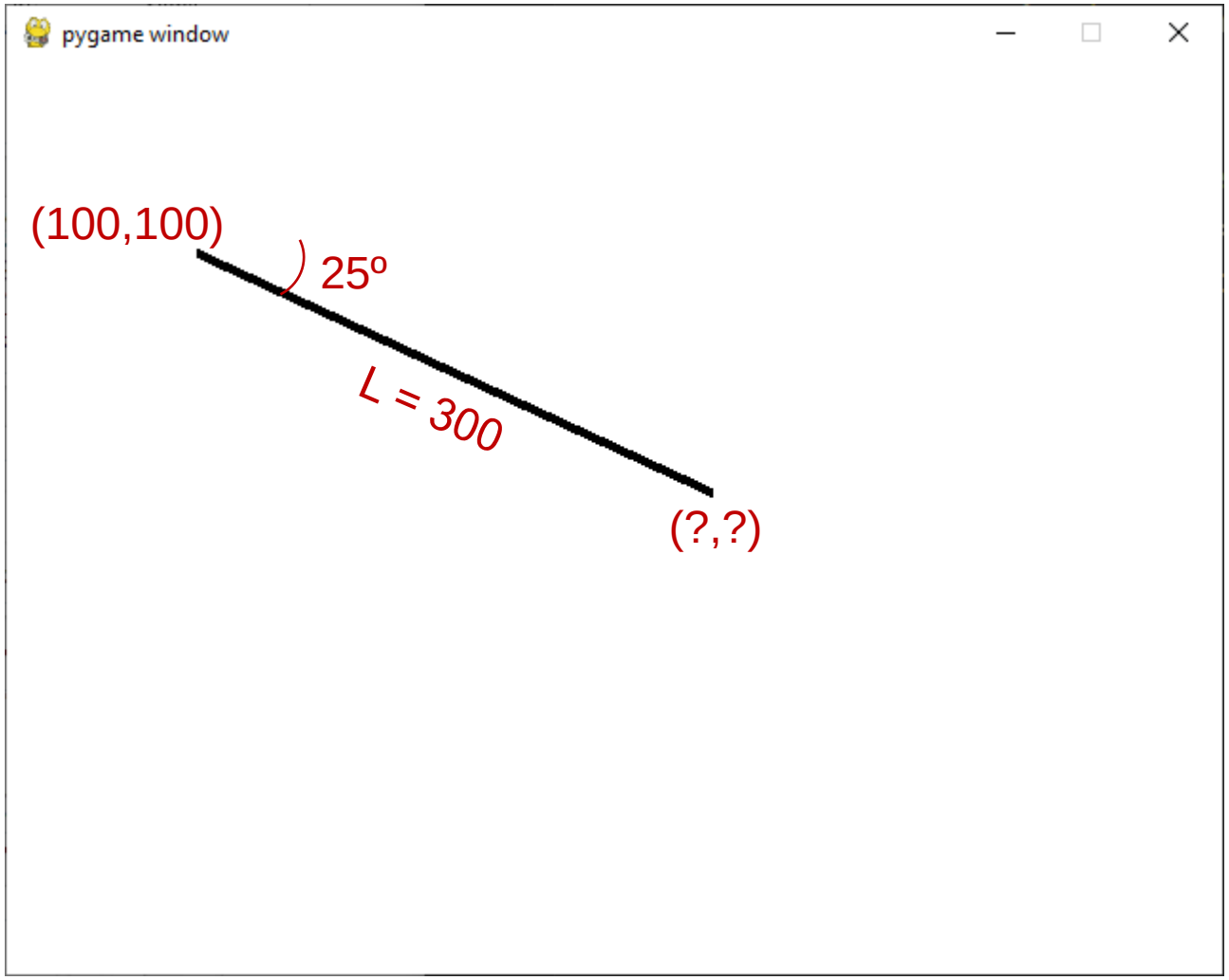


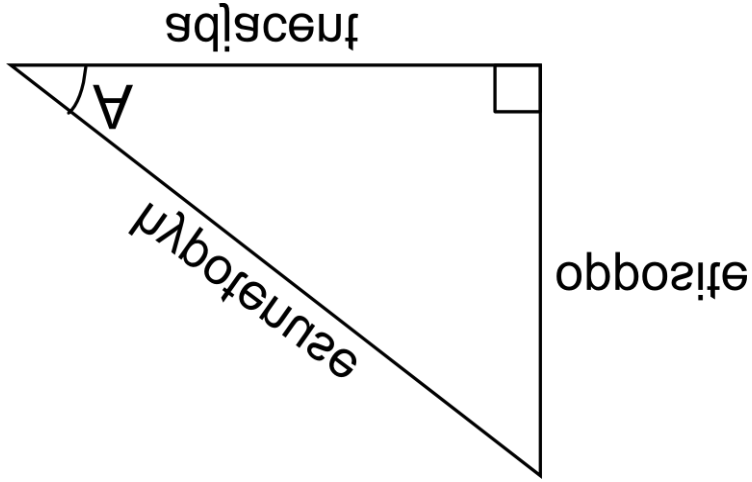
Drawing a line at an angle





Drawing a line at an angle





- $\sin(A) = \frac{opposite}{hypotenuse}$
- $\cos(A) = \frac{adjacent}{hypotenuse}$

Exercise time!

